

- [5]

- The mean height of 10 soft rush plants was calculated at each of eight different altitudes.

Table 8.1

altitude/m	mean height of soft rush plants/cm
100	85
150	86
200	83
250	79
300	72
350	74
400	68
450	63

The statistical test, Spearman's rank correlation (r_s), was applied to find out if there was a relationship between the altitude and the mean height of the soft rush plants.

(i) The formula for calculating Spearman's rank correlation is:

$$r_s = 1 - \left(\frac{6 \times \Sigma D^2}{n^3 - n} \right)$$

- ΣD^2 is the sum of the differences between the ranks of the two samples
- n is the number of samples.

In this investigation the value of ΣD^2 is 164.

Calculate the value of r_s .

Show your working and write your answer to **two** decimal places.

$r_s =$ [2]

(ii) Use your value for r_s to evaluate the relationship between the altitude and the mean height of the soft rush plants.

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 [2]

[Total: 9]

Section B

Answer **one** question.